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SUPERSEDING
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MILITARY SPECIFICATION

CLOTH, INSULATION, MULTIPLE LAYER COMPOSITE, QUILTED, FLAME RESISTANT

1. SCOPE

1.1 Scope. This document covers a quilted composite insulation cloth made up of layers of coated glass fabric, flame resistant polyester batting, and metallized film.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-T-285	-	Thread, Polyester
PPP-P-1136	-	Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA, 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 518 - Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter
D 578 - Glass Fiber Yarns
D 579 - Greige Woven Glass Fabrics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The finished insulation cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.5).

3.3 Materials and components.

3.3.1 Cover fabric. The coated cover fabrics shall conform to the requirements of table I when tested as specified in 4.4.3.

TABLE I. Physical requirements of coated cover fabrics

Characteristics	Requirements	
	Impermeable	Permeable
Weight, finished, oz/yd^2(max)	5.5	5.5
Width, finished, in.	54.0 ± 1.0	54.0 ± 1.0
Yarns per inch (min):		
Warp	52	52
Filling	46	46
Breaking strength, lbs. (min):		
Warp	95	85
Filling	95	85
Creased breaking strength; lbs.(min):		
Warp	18	12
Filling	24	18
Air permeability, ft^3/min/ft^2	1.0 (max)	5.5 (min)

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TABLE I. Physical requirements of coated cover fabrics (cont'd)

Characteristics	Requirements	
	Impermeable	Permeable
Spray rating	90	90
Hydrostatic pressure, cm (min):		
Initial	75	22
After low temperature exposure	1/	1/
Flame resistance:		
After flame, seconds (max)	2	2
Percent consumed, percent (max)	30	30
Flexibility, in-lb (max):		
Initial	0.010	0.010
Low temperature	0.040	0.040
Topstitched seam efficiency, percent (min)	80	80
Abrasion resistance, number of cycles (min)	350	350
Low temperature resistance	2/	2/

1/ Not more than three separate droplets of water shall appear at the crease of the specimen under a hydrostatic head of less than 50 cm for impermeable coated cloth, or at less than 20 cm for the permeable coated cloth.

2/ The specimen shall not crack.

3.3.1.1 Base cloth. The base cloth shall meet the requirements of ASTM D 579 except it shall consist of either ECC 150 1/0 or ECD 225 1/0 fiberglass yarns as designated in ASTM D 578, in a plain weave construction with a maximum weight of 4 ounces per square yard.

3.3.1.2 Coating compound. The coating compound shall be an approved acrylic or a urethane-acrylic aqueous polymeric system which is coreactive with water repellent silicons and flame retardant elements (see 6.4). The compound shall be compatable with the base cloth to insure it's proper adherence to the yarns of the base cloth.

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3.3.1.2.1 Impermeable coating. The coating compound shall be applied to the base cloth for the inner panel to yield an impermeable coating which conforms to the requirements in table I. The face of the coated cloth shall be marked "FACE IMPERMEABLE".

3.3.1.2.2 Permeable coating. The coating compound shall be applied to the base cloth for the outer panel to yield a permeable coating which conforms to the requirements in table I. The face of the cloth shall be marked "FACE PERMEABLE".

3.3.1.2.3 Color. The finished cloth shall be natural and shall be equal to or whiter than the standard sample.

3.3.2 Flame resistant polyester batting.

3.3.2.1 Fiber. The fiber used in fabricating the batting shall be made of inherently flame resistant virgin polyester manufactured from polyethylene teraphthalate resin modified by phosphorous compounds which are reacted and incorporated into the polyethylene teraphthalate polymer chain. These fibers shall not support combustion. The batting fiber blend shall be 75 percent of 6 denier, 2-1/4 ± 1/4 inch staple length fibers with 10±1 crimps per inch and 25 percent of 15 denier, 3 ± 1/8 inch staple length fibers with 9±1 crimps per inch. The fibers shall have a minimum melting point of 478°F with a water soluble antistatic finish not exceeding 1 percent on fiber weight (see 4.4.2).

3.3.2.2 Bonding agent and application. The flame resistant polyester fibers specified in 3.3.2.1 shall be bonded with an approved durable flame retardant bonding agent (see 6.4) uniformly applied to both sides. The concentration of the applied bonding agent shall be 18 ± 2 percent on the weight of the finished bonded cut staple batting (see 4.4.2).

3.3.2.3 Finished bonded batting. The bonded fibers in the finished cut staple flame resistant batting shall be well opened, uniform, and separated without excessive breakage. The batting shall show no more neps and multiple fiber ends than the standard sample (see 6.3). The finished batting shall conform to the physical and thermal properties listed in table II when tested as specified in 4.4.3.

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TABLE II. Physical and thermal properties of flame resistant polyester batting

Characteristics	Requirement
Weight, oz/yd^2	2.0 ± 0.2
Width, in.	54.0 ± 1.0
Thickness, in. @ 0.01 psi (min)	0.30
Batting evenness, oz/yd^2	± 0.3 1/

1/ The evenness of the batting shall be such that no single weight determination deviates more than 0.3 oz/yd^2 from the mean of five specimens tested as specified in 4.4.3.

3.3.3 Metallized reflective film. The metallized reflective film base shall be 75 gage, low density polyethylene film base with 99.9 percent pure aluminum vapor deposited on the base having a surface resistivity of 2 to 3 ohms per square prior to top coating (see 4.4.2). The metallized film shall be topcoated with an approved electron beam curable, acrylate epoxy coating to provide abrasion resistance for the aluminum coating (see 6.4). The film shall conform to the physical properties listed in table III when tested as specified in 4.4.3.

TABLE III. Physical requirements of finished metallized reflective film

Characteristic	Requirement
Weight, oz/yd^2	0.5 ± 0.1
Width, finished, in.	54.0 ± 1.0
Thickness, mils	0.75 ± 0.05

3.3.4 Thread. The thread used for the needle and bobbin in the quilting fabrication of the multiple layer composite shall conform to type I, class 1, subclass B, size E/3 ply of V-T-285.

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3.3.4.1 Thread color. Two thread colors shall be used for identification of the two panels which makeup the total composite. The thread color for the quilting of the outer permeable panel (see figure 3) shall be Olive Drab S-1, C.A. 66022. The thread color for the quilting of the inner impermeable panel (see figure 3) shall be natural.

3.4 Quilted composite.

3.4.1 Panels. The inner and outer panels shall be separately fabricated as shown in figure 1 using the zigzag pattern of quilting stitching shown in figure 2 such as to conform to figure 3.

3.4.2 Cloth alignment. The coated fiberglass cover fabrics (3.3.1.2.1 and 3.3.1.2.2) shall be orientated in the quilted panels so the identified fabric face is on the exterior facing outward. The cloth covering, layers of polyester batting and metallic film for each panel shall be positioned as to yield a straight edge on the side of the ensemble. The batting and film components shall be flush or extend beyond the cloth covering selvages on both sides of the ensemble. Any extension of the materials on the straight edge shall not exceed 1/4 inch wide while the extension of materials on the opposite side shall not exceed 1 inch. The extending edges of the filler materials shall be evenly trimmed without ragged edges.

3.4.3 Stitching and stitch pattern. All stitching shall conform to type 301 or type 401 of FED-STD-751. The quilt pattern shall be a zigzag pattern conforming to figure 2 with 4 to 6 stitches per linear inch in the 6-inch pattern repeat. The stitch pattern for each panel shall be evenly offset such that stitching rows do not overlap when the two panels are placed together and used as one composite (figure 3). The distance between zigzag rows shall be 10 $\pm$ 1/4 inches. All panel component layers shall be sewn together at the selvage area to facilitate handling the multiple component layers in the end item edge binding and seaming operations. Two color threads shall be used in stitching for identification purposes. The stitching on the permeable outer panel shall be Olive Drab S-1, while the quilt stitch on the inner impermeable panel shall be natural.

3.4.3.1 Thread breaks and open stitching. Thread breaks and open stitches shall be secured by stitching back of the break in conformance with the pattern, not less than 1 inch.

3.4.4 The composite configuration. The inner impermeable and outer permeable panels of this composite shall be placed batting sides together and must be utilized as one composite. The composite of outer cloths, batting layers and metallic film shall conform to the requirements of table IV, when tested as specified in 4.4.5.

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TABLE IV. Physical and thermal properties of the finished quilted composite

Characteristic	Requirement
Composite weight: oz/yd^2. (max)	22.0
Width, finished: in.	54.0 ± 1.0
Composite thickness: in. @ 0.002 psi (max)	2.0
Composite R value: $R=^oF \cdot Ft^2 \cdot H \cdot Btu^{-1}$	3.2
Flame resistance, after flame: sec. (max)	6.0

3.5 Length and put-up. All quilted composites shall be furnished without fabric splices (joining of fabric outer coverings) and without laps or gaps in the batting and film fillers. Unless otherwise specified (see 6.2), quilted multiple layer composites, purchased directly by the Government, shall be furnished in continuous lengths of 61 ± 1 yards. Each length shall be put-up on a roll as specified in 5.1.

3.6 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.7 Workmanship. The quilted multiple layer composite shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.4.1 and tested for the characteristics specified in table VIII. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 Certification. Components listed in table V shall be inspected on the basis of a contractor's certificate of compliance with the indicated requirements.

TABLE V. Component certification

Component	Characteristic	Requirement paragraph
Base cloth	Yarn, weight; weave, yarn count	3.3.1.1

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TABLE V. Component certification (cont'd)

Component	Characteristic	Requirement paragraph
Flame resistant polyester fiber	Material identification 1/, crimp, denier, length of staple, melting point, anti-static finish	3.3.2.1
Bonding agent	Resin trade name Percent on bonded batting	3.3.2.2
Metallic film	Percentage pure aluminum Surface resistivity	3.3.3

1/ The certification shall include fiber manufacturer's identification data.

4.4.3 Component testing. In addition to the quality assurance provisions of referenced documents, testing shall be performed on components and materials listed in table VI for the characteristics shown. Test methods cited shall be in accordance with FED-STD-191 except where otherwise specified. The sample unit for the permeable coated base cloth, the impermeable coated base cloth, the polyester batting and the reflective film shall be 3 continuous yards each, full width. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VI. Component tests

Component	Characteristic	Requirement paragraph	Test method
Coated cover fabric	Weight	3.3.1	5041
	Width	3.3.1	5020 1/
	Yarns per inch	3.3.1	5050
	Breaking strength	3.3.1	5100
	Creased breaking strength	3.3.1	4.5.3, 5100
	Air permeability	3.3.1	5450
	Spray rating	3.3.1	5526

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TABLE VI. Component tests (cont'd)

Component	Characteristic	Requirement paragraph	Test method
	Hydrostatic pressure After low temperature exposure	3.3.1	4.5.5
	Flame resistance:		
	After flame	3.3.1	5905
	Percent consumed	3.3.1	5905
	Flexibility, initial and at low temperature	3.3.1	5202 2/
	Topstitch seam efficiency	3.3.1	5110 3/
	Abrasion resistance	3.3.1	5302 4/
	Low temperature resistance	3.3.1	4.5.4
Flame resistant batting	Weight	3.3.2.3	5041
	Thickness	3.3.2.3	5030 5/
	Evenness	3.3.2.3	5041 6/
Metallized film	Weight	3.3.3	5041
	Thickness	3.3.3	5030

1/ Method 5020, short specimen method.

2/ Except that five specimens shall be tested in each of the warp and filling directions. The specimens shall be evaluated initially at standard conditions. New specimens shall be prepared for low temperature flexibility and both the specimens and the instrument shall be conditioned at minus 20° ± 5°F for at least 1 hour and tested at that temperature as specified in Method 5202.

3/ Method 5110 except each test specimen shall be a single layer of fabric prepared by topstitching with stitch type 301, 8 inches from the specimen edge parallel to the warp yarns. Topstitching shall be 7 to 9 stitches per inch with thread conforming to V-T-285, type I, class 1, subclass B, size E/3 ply and shall utilize a needle measuring a minimum of 0.044 ± 0.001 inch across the blade at the eye. For the purposes of this test, the topstitching shall be considered a seam.

4/ Method 5302 utilizing grit 0 emery paper abradent, air pressure under the diaphragm controlled at 3 pounds per square inch, 2-1/2 pound vertical pressure, and a multidirectional rubbing action run to a state of destruction.

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5/ Except that the specimens shall be measured at a load of 0.002 psi.

6/ Except that the results shall be reported for each of the five specimens as the deviation from the mean of the weights of the five specimens.

4.4.4 End item examination.

4.4.4.1 Yard-by-yard examination. The cloth shall be examined on both sides for the defects listed below. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The lot size shall be expressed in yards. The sample unit shall be 1 linear yard. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects. The number of rolls from which the sample yardage is to be selected shall be in accordance with table VII. The sample yardage shall be apportioned equally among the selected rolls.

Examine	Defect	Classification Major	Minor
Cover fabrics	Material not as specified	X	
	Color not as specified	X	
	Any smash	X	
	Any cut, hole, or tear	X	
	Spot or stain, clearly visible 1/		X
	Abrasion resulting in a weak place or any weak place	X	
	Floats and skips:		
	- Multiple, 3/4 inch or more in combined warp and filling directions	X	
	- Multiple, less than 3/4 inch in length in combined warp and filling directions or single float or skip extending over more than one warp-wise or filling-wise inch		X

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Examine	Defect	Classification Major	Minor
Cover fabrics (cont'd)	Broken or missing yarns (warp and filling):		
	- Three or more contiguous, regardless of length	X	
	- Two or more contiguous missing for 1 inch or more	X	
	- Two or more contiguous missing for less than 1 inch		X
	Single yarn missing for 4 inches or more	X	
	Crease or wrinkle, embedded	X	
	Five or more kinks, knots, or loops in 9 linear inches, clearly visible 1/ and protruding from surface of fabric		X
	Any tight warp section resulting in waviness or dimensional distortion of fabric	X	
	Any cut, broken, torn, folded or rolled selvage	X	
	Any stringy or loopy, scalloped, tight, or slack selvage		X
Thread, polyester	Not color specified or not within established tolerances		X
Workmanship	Stitching pattern not within established tolerances by 1/4 inch or less		X
	Stitching pattern not within established tolerances by more than 1/4 inch	X	
	Thread ends not trimmed		X
	One or more rows of stitching omitted	X	
	Needle chews	X	
	Loose or irregular tension		X
	Stitch type not as specified	X	
	Broken or missing stitches 1 inch or more (not repaired)	X	
	Broken or missing stitches less than 1 inch (not repaired)		X

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Examine	Defect	Classification Major	Minor
	Average stitches per linear inch on 6-inch repeat, less than four or more than six	X	
	Repaired thread breaks or open stitches stitched back less than 1 inch		X
	Face side of coated fiberglass cover fabric not orientated to exterior	X	

1/ At normal inspection distance of approximately 3 feet when viewed against a black background.

4.4.4.2 Overall examination. The cloth shall be examined for the defects listed below. Each defect listed shall be counted not more than once in each roll examined. The sample size shall be the applicable number of rolls indicated in table VII. Each roll in the sample shall be examined over its entire length. The lot shall be rejected if the total number of defects in the sample exceeds the acceptance number specified in table VII.

- Overall uncleanness.
- Objectionable odor.
- Edges of bonded batting not evenly trimmed.
- One edge not straight.
- Batting recessed or extends more than 1/4 inch on straight edge or batting on opposite side extends more than 1 inch from cloth selvages.
- Constructed batting exhibits overall uneveness or contains cloth distortions or bulges.
- Width not as specified.
- Lapping or otherwise joining of ends of batting or splicing (joining of ends of cover cloths) to make a continuous length.
- Gaps or bare area in polyester batting material.

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TABLE VII. Sampling and acceptance criteria

Lot size in yards	Sample size in rolls	Acceptance number
Up to and including 1,300 1/	3	0
1,301 up to and including 3,200	5	0
3,201 up to and including 8,000	7	0
8,001 up to and including 22,000	10	0
22,001 up to and including 110,000	15	0
110,001 and over	25	1

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.4.4.3 Length examination. The sample for length examination shall consist of those rolls selected for overall examination in accordance with 4.4.4.2. Each roll in the sample shall be examined for continuous length. Any continuous length found to be less than the specified minimum or more than the specified maximum shall be scored as a defect. The acceptance number shall be as specified in table VII. In addition, the lot shall be rejected if the total of actual lengths of the rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.5 End item testing. The end item shall be tested for the characteristics indicated in table VIII. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of 1 yard. The sample unit shall be 5 yards of the finished quilted composite. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (no. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

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TABLE VIII. End item tests

Characteristic	Requirement paragraph	Test method	No. of determinations sample unit	Results reported to nearest
Weight	3.4.4	5041	5	0.1 oz./yd.^{2}
Thickness	3.4.4	5030 1/	5	0.1 in.
R Value	3.4.4	4.5.2	2	0.1 $^oF \cdot Ft^2 \cdot H \cdot Btu^{-1}$
Flame resistance, after flame	3.4.4	4.5.1	5	1 second

1/ Except the load shall be 0.002 psi.

4.4.6 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

4.5 Methods of inspection.

4.5.1 Flame resistance test.

4.5.1.1 Preparation of specimens. Cut a specimen from each outer permeable and inner impermeable quilted panel with dimensions of 26 inches wide by 13 inches long, such that the outer permeable panel quilt line is approximately 10 inches in from the left side and the inner impermeable panel quilt line is 5 inches in from the left. Place panels batting sides together, lining up left edge even. Stitch sides and top of specimen. On outer panel side, mark at 2-3/8 inches and 2-5/8 inches from quilting stitch (the warp direction) across the full width of the specimen. Stitch along both markings to yield parallel stitch lines. Stitch the specimen across the bottom. Cut between parallel stitching lines to yield five test specimens, each approximately 5 inches wide by 12 inches long.

4.5.1.2 Procedure. Evaluate flammability in accordance with FED-STD-191 Method 5905 except that the specimen shall be subjected to only one 12-second immersion into the flame. The specimens shall be evaluated in the warp direction only.

4.5.1.3 Results. Average the after flame results of the five specimens and report.

4.5.2 R value test.

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4.5.2.1 Preparation of specimen. The dimensions of the test specimens shall be dictated by the heat flow meter. When evaluating the quilted composite, the total composite shall be made up of the inner permeable and outer impermeable panel and shall be evaluated with batting sides together. The test specimen shall be sampled such that a quilted portion of the composite shall be included in the metering area of the heat flow meter.

4.5.2.2 Procedure. The test specimen shall be evaluated in accordance with ASTM C 518 with the hot plate (T_h) at 120°F and the cold plate (T_c) at 72°F. The hot and cold plates shall contact the specimen at a static load of 0.002 pounds per square inch.

4.5.2.3 Results. The results shall be reported in thermal resistance units of $^{o}F \cdot Ft^{2} \cdot H \cdot Btu^{-1}$ which is expressed as "R" value.

4.5.3 Breaking strength after creasing. Twenty 8 by 4-inch specimens shall be cut from each sample unit. Ten specimens shall be cut with the long dimension in the warp direction and ten with the long dimension in the filling direction. No two specimens shall contain the same warp yarns for the warp test nor the same filling yarns for the filling test. No specimens shall be taken nearer the selvages than 1/10 of the fabric width. Loop each specimen end to end with the back sides of the fabric together. Do not flatten the loop by hand. Each specimen shall be creased by rolling a steel cylinderical 5-pound weight parallel to and centered over the loop. Push the weight so that it rolls smoothly and straight over the looped area without applying any pressure other than the weight of the roll. Roll the weight at a rate in which it will transverse the specimen in approximately 1 second. A single roll across the specimen will be one pass and shall be repeated until five passes have been applied. The creased specimen shall be tested for breaking strength in accordance with Method 5100 of FED-STD-191, except the specimen shall be 8 by 4 inches and the pulling speed will be 12 inches/minute using a CRE apparatus with rubber faced jaws. Unfold the creased specimen, being sure not to fold backwards beyond 180 degrees. Position the creased area midway between the clamps of the tensile testing apparatus and test until the specimen breaks. The creased breaking strength shall be the average of the results obtained from the specimens tested in the warp and filling directions and shall be reported separately.

4.5.4 Low temperature resistance. Method 5874 of FED-STD-191 shall be used except that five specimens shall be tested from each sample unit, the specimens shall be exposed to - 20° ± 5°F for not less than 4 hours, and the weight of the steel roller shall be 5 pounds.

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4.5.5 Hydrostatic resistance after low temperature exposure. This test shall be conducted only for lots meeting the low temperature resistance requirements in table I. The same specimens tested in accordance with 4.5.4 shall be tested for hydrostatic resistance in accordance with Method 5514 of FED-STD-191. The water shall be applied to the face side and creased area of the specimen. The height of the column at which any specimen fails shall be reported to the nearest 0.5 cm. The lot shall be rejected if any specimen fails to meet the specified requirement.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Levels A and Commercial. The multiple layer cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial. The multiple layer cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

6.1 Intended use. The quilted composite consisting of the inner impermeable and outer permeable panels assembled batting sides together, is intended for use in the manufacture of insulated tent liners.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.2, 4.3, and 6.5).
c. Length, if other than specified (see 3.5).
d. Selection of applicable levels of put-up, preservation, and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

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6.4 Bonding agents and coatings. Approval of bonding agents for use in battings and coatings for use in the metallized film and glass fabric covered by this document, is the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 07160-5014. Approval is based on extensive evaluations, including those for toxicity, which are not set forth in the document. Because of the time necessary to conduct a full evaluation, only those bonding agents and coatings already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of the first article.

6.6 Subject term (key word) listing.

Cloth, insulation, quilted
Flame resistant
Tent liners

6.7 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8305-0091

Review activities:

Army - MD
Air Force - 82
DLA - CT

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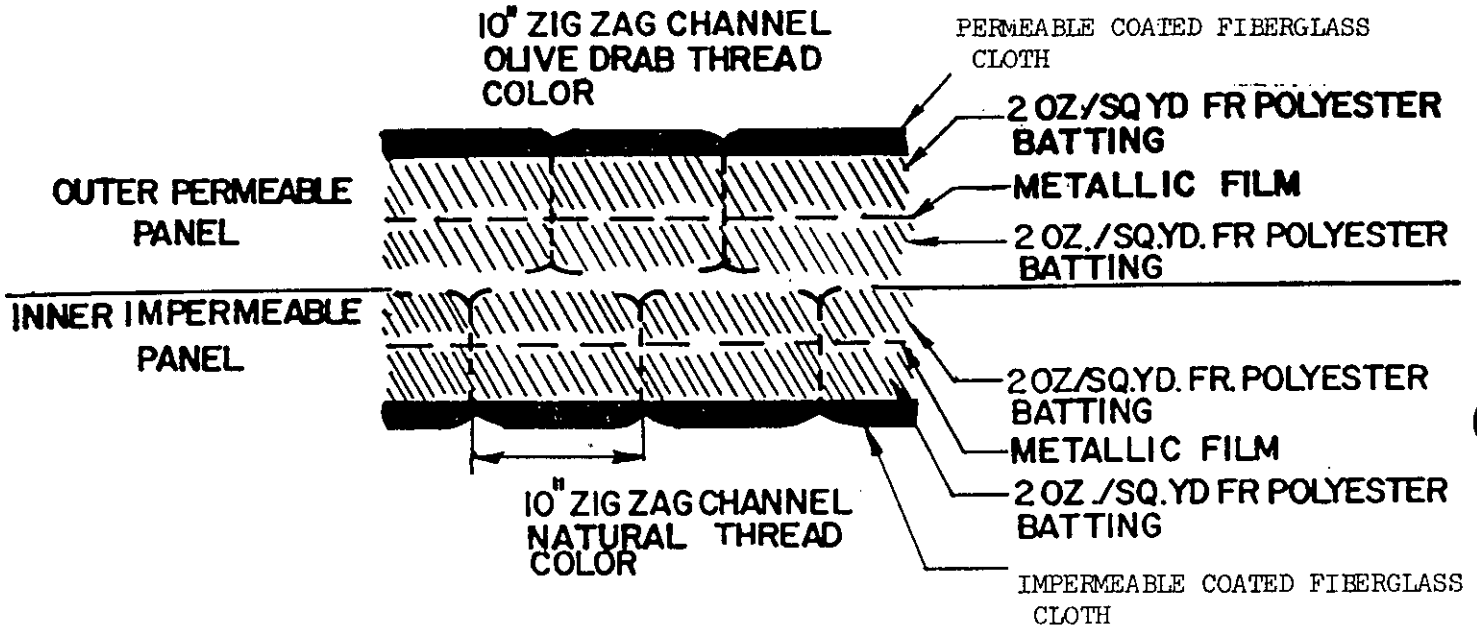


FIGURE 1 :

SCHEMATIC CROSS SEC. OF CLOTH, INSULATION, MULTIPLE LAYER COMPOSITE, QUILTED, FLAME RESISTANT

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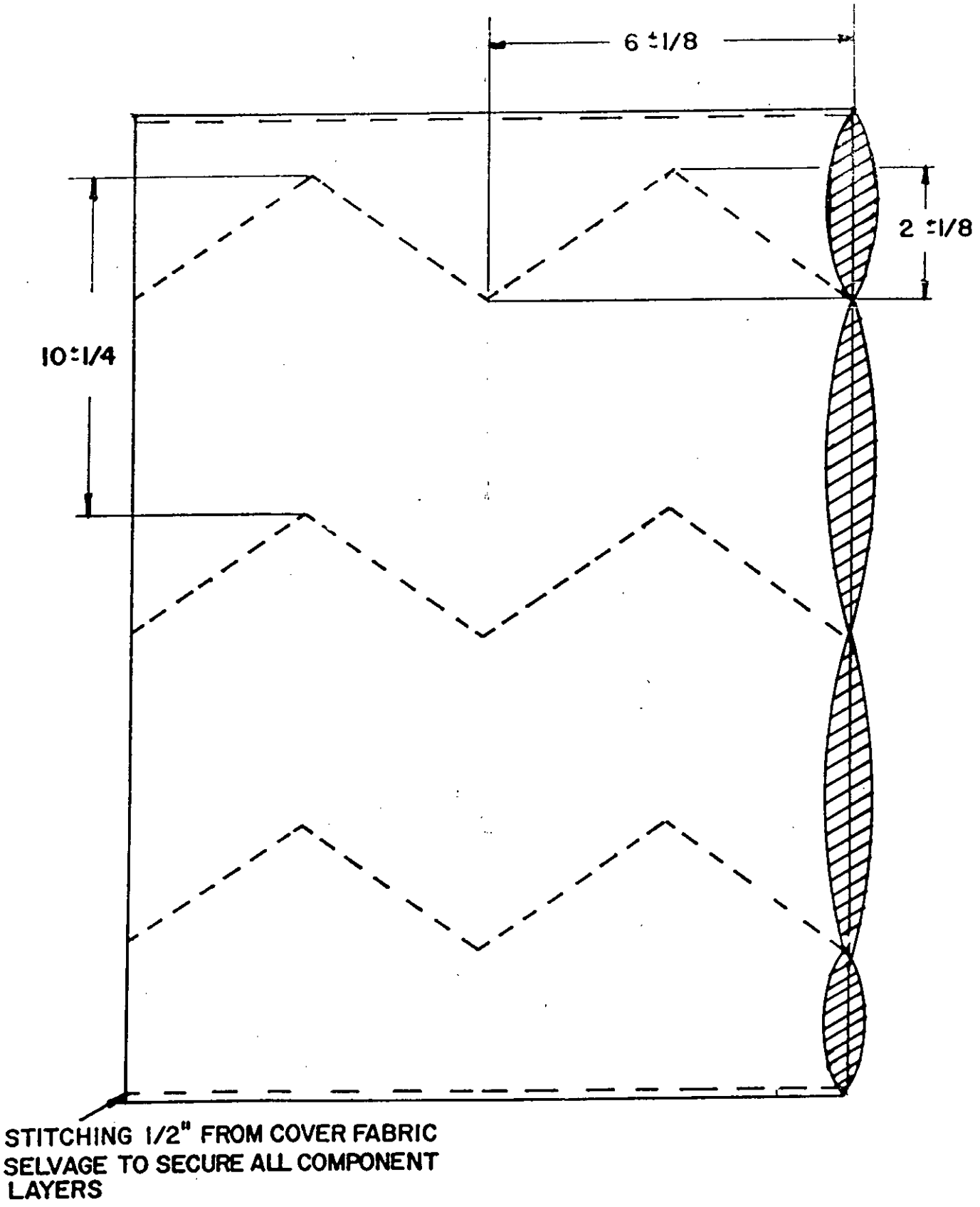


FIGURE 2: ZIG ZAG PATTERN

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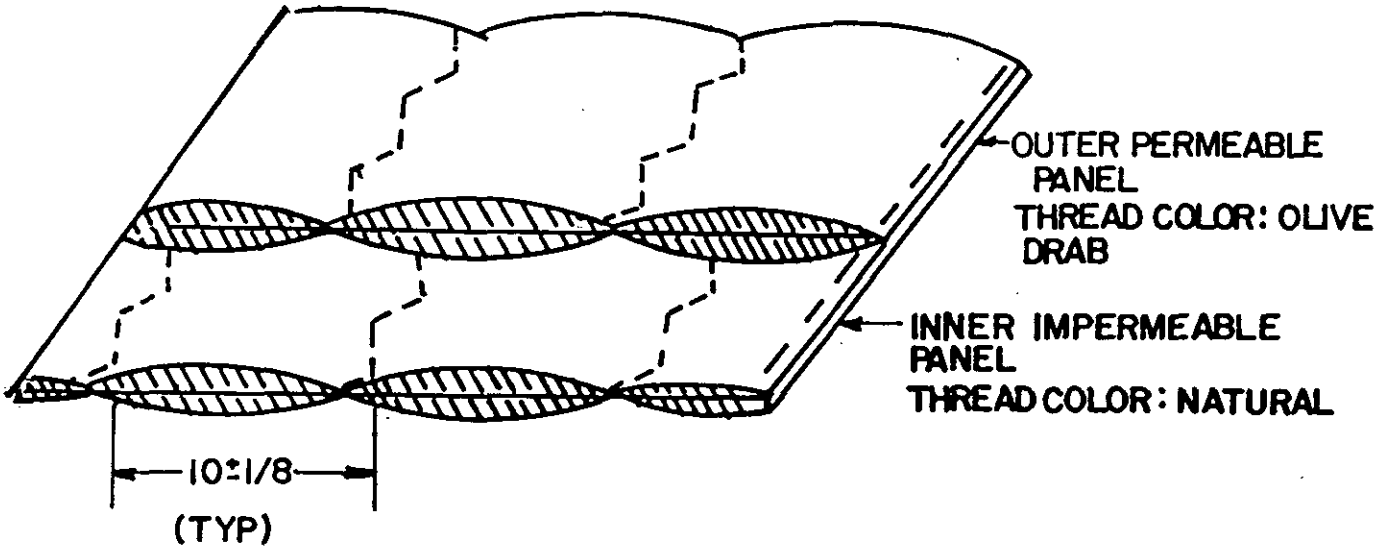


FIGURE 3: QUILT CONST., ZIG ZAG PATTERN, OFFSET QUILT STITCHING ROWS

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44154A	Cloth, Insulation, Multiple Layer Composite, Quilted, Flame Resistant

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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